

Review

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Topics for today

1. Heat transfer
2. Mass transfer
3. Model formulation
4. VS Code

How we'll do it

- Instructor presentation of solution
- In-class problem
- Mixing up groups
- Menti poll
- Student presentation of solution

Melting ice model (class 4)

Think about *model formulation steps*

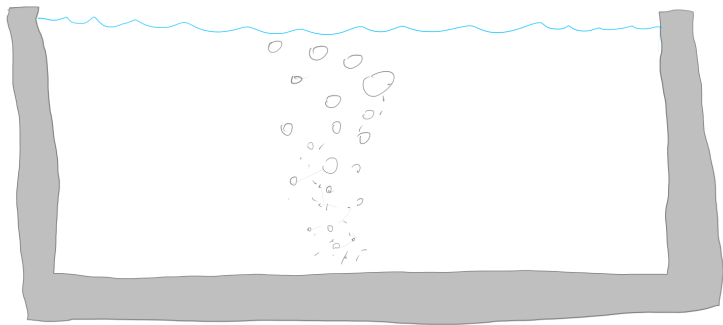
Think about *modules* and *scripts*

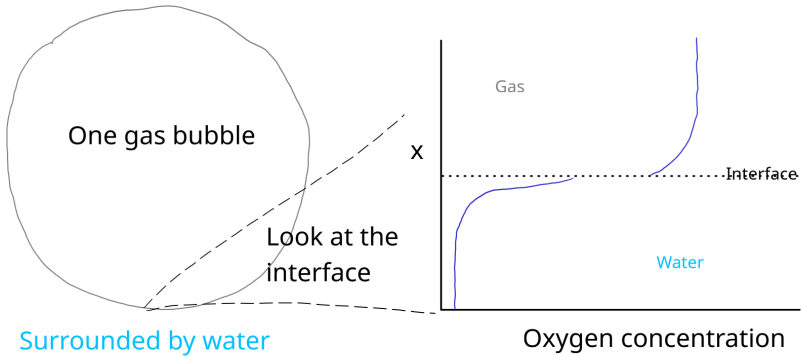
Heat and mass transfer modes

Tricky topic!

Key concepts

- Discontinuity exists at interface
- Profiles unknown and unimportant
- Consistent units and phases needed for flux calculation
- Think about equilibrium
- Mass transfer coefficients k_c are usually based on *measurements*





In-class problem

This one has some overlap with the benzene problem.

A thin layer of animal slurry is spread on an agricultural field for use as fertilizer. Given a free ammonia (NH_3 (aq)) concentration of 10 mg/L and a mass transfer coefficient of 0.01 m/s, what is the initial rate of ammonia volatilization (flux)? What is the initial mass flow rate if the slurry has a surface area of 1 ha (10,000 square meters)? What would happen to the liquid-side profile and flux over time if

- the slurry layer is very still, with no mixing, or
- the slurry layer is well-mixed below a thin film at the interface?

In-class problem

Given a total ammonia concentration of 3 g/L in the slurry, would you expect that initial flux to persist for days?

If all liquid-side resistance to transport is in a 100 micron layer at the slurry surface, is it reasonable to assume steady-state within that film?

Discuss, calculate, draw, ...

Student presentation of results

Work on Chapter 5 problems

Try to complete problem 2

Ask for help!

1. Fish gills
2. Benzene volatilization

Chemical reactions and microbiology

Read Chapter 6!